



Immunological Aspects of the GIT

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which cell type samples antigen across the epithelium, and which detects helminths?
2. Where is secretory IgA made, and what three things does it do in the lumen?
3. How does food intolerance differ from food allergy in mechanism?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The gut immune balancing act

- The GIT is both a digestive organ and a **gateway** for pathogens, toxins and dietary antigens
- It balances **inflammation vs suppression, active immunity vs tolerance, protection vs autoimmunity / allergy** — failure of any balance causes disease
- Immune failure underlies **IBD, IBS, food allergy and GI infections**; **innate** immunity = first line; **adaptive** (B and T cells) = memory

2 Innate defences — and what happens when they fail

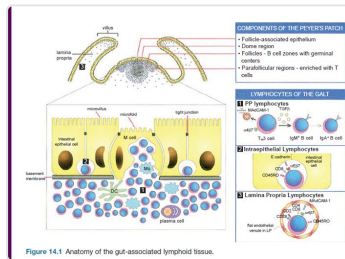
Defence	Role
Tight junctions	Seal epithelial cells
Saliva and pancreas	Lysozyme ; pancreatic antimicrobial peptides
Stomach acid	Kills ingested microbes
Mucus + antimicrobial factors	Made by intestinal mucosa
Bile acids	Immune modulators: signal via receptors on macrophages / DCs to keep liver and gut tolerogenic
Microbiota and metabolites	Colonisation resistance
Epithelial turnover + motility	Purge harmful microbes
If this fails...	...this results
Achlorhydria (atrophic gastritis)	Bacterial overgrowth
Altered flora (antibiotics)	Pathogens flourish — <i>C. difficile</i> pseudomembranous colitis
Disordered motility	Small-intestinal bacterial overgrowth , permeability, translocation
Tight-junction disturbance	" Leaky gut " → systemic inflammation, e.g. IBD
Bile-acid imbalance	Implicated in Crohn's disease and ulcerative colitis

3 GALT — where antigen is sampled

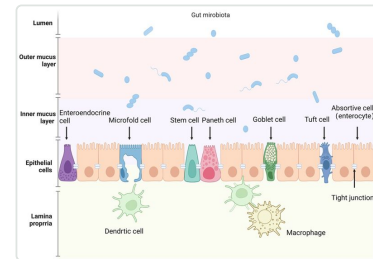
Structure	Key point
Peyer's patches	Lymphoid follicles of the small intestine, densest toward the terminal ileum ; trap antigens; site where B and T cells meet antigen
Appendix	Bona fide adaptive immune inductive site, structure like Peyer's patches

Isolated lymphoid follicles (ILFs)

Along the whole intestine; **change quickly with microbiome and cytokines**; a primary source of **IgA-secreting B cells**



Anatomy of the gut-associated lymphoid tissue.



Epithelial cell types and mucus layers.

4 The epithelial cast — one job each

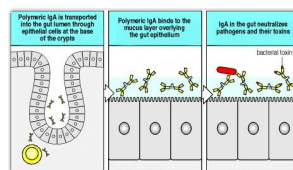
Cell	Job / clinical hook
Enterocyte	Absorptive; expresses TLR4 internally; defects → congenital enteropathies
Enteroendocrine	Senses microbial metabolites; releases cytokines and hormones (e.g. serotonin)
Goblet	Mucins → mucus layer; secretes AMPs
Paneth	α-defensins, lysozyme ; abnormalities → Crohn's disease
M cell	Transcytosis of antigen across the epithelium to macrophages, DCs and lymphocytes
Tuft cell	Detects helminths and starts a type 2 immune response

5 Lamina propria and secretory IgA

- Lamina propria** holds most immune cells: APCs, mast cells, ILCs, **T cells (CD4, CD8, Th17, Treg)** and **IgA plasma cells**; Th1 = cell-mediated (intracellular pathogens); Th2 = antibody-mediated (allergens, parasites)
- Homeostasis is dominated by **IL-10 and TGF-β**; **ILC3 / Th17** make **IL-22** → AMPs (but can also drive inflammation); eosinophils help in parasitic and allergic responses



sIgA is the most abundant antibody in the intestinal lumen; it neutralises toxins (e.g. cholera toxin), viruses and bacteria, blocks adherence, and blocks antigen entry into the systemic immune system.



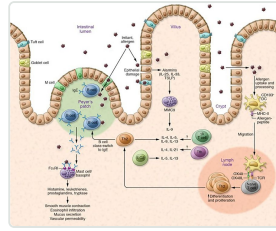
Polymeric IgA transport and function.

6 Food intolerance vs food allergy — discriminator grid

	Food intolerance	Food allergy
Mechanism	Non-immune: enzymatic, pharmacological or metabolic	Immune-mediated: IgE (Th2), non-IgE, or mixed
Example	Lactose intolerance (no lactase → cannot split lactose to glucose + galactose); fructose, histamine	Peanuts, milk, eggs, fish
Organs	GI only — profuse diarrhea or vomiting; no respiratory or skin involvement	Skin, GI mucosa, respiratory — nausea, cramps, vomiting, watery / mucous diarrhea; may progress to anaphylaxis
Diagnosis	—	Difficult; skin-prick tests have limited predictive value; elimination diets

Treatment	Food avoidance	Education, strict elimination ; antihistamines / immunotherapy in some (e.g. peanut); anaphylaxis = emergency
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Mechanism of food allergy: Th2 cytokines (IL-4, IL-5, IL-13, IL-9, IL-25, IL-33) → B-cell class switching → **IgE** → mast cell / basophil degranulation → smooth-muscle contraction, eosinophil infiltration, mucus secretion, increased vascular permeability.



Mechanism of allergic reaction in food allergy.

P Must-remember pearls

- **M cells** transport antigen; **Paneth cells** make defensins and lysozyme; **tuft cells** sense helminths → type 2 response.
- **Peyer's patches, appendix and ILFs** are GALT; ILFs are a main source of **IgA B cells**.
- **sIgA** = made by lamina-propria plasma cells, transported across epithelium; **neutralises, blocks adherence, blocks antigen uptake**.
- Homeostasis relies on **IL-10 and TGF-β**; ILC3 / Th17 → **IL-22** → **AMPs**.
- **Intolerance = non-immune** (lactase); **allergy = immune** (IgE, Th2; skin + GI + respiratory; anaphylaxis).

Q Test yourself

1. A patient with atrophic gastritis and achlorhydria develops small-bowel bacterial overgrowth. Which innate defence has failed?

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|--------------------------|--------------------|
| A. Gastric acidity | B. Tight junctions |
| C. Peyer's patch density | D. Sweat lysozyme |

2. A patient has a parasite infection with a type 2 response starting at the epithelium. Which cell first detects the helminth?

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|----------------|---------------|
| A. Goblet cell | B. Tuft cell |
| C. Paneth cell | D. Enterocyte |

3. After a meal of milk, an adult has diarrhea and vomiting but no rash, wheeze or hives. Most likely diagnosis?

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|------------------------------|------------------------|
| A. IgE-mediated milk allergy | B. Lactose intolerance |
| C. Anaphylaxis | D. Celiac disease |

4. Which best describes the function of secretory IgA in the lumen?

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|---------------------------------|---|
| A. It activates mast cells | B. It neutralises toxins and blocks microbial adherence |
| C. It causes tissue destruction | D. It presents antigens to T cells |

5. Antibiotic use allows toxin-producing bacteria to overgrow and cause pseudomembranous colitis. Which organism?

- | | |
|------------------------------------|---------------------------|
| A. <i>Clostridioides difficile</i> | B. <i>Bifidobacterium</i> |
| C. <i>Lactobacillus</i> | D. <i>H. pylori</i> |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Loss of gastric acid allows bacterial overgrowth.
2 — B — Tuft cells detect helminths and initiate the type 2 response.
3 — B — Intolerance is non-immune and limited to GI symptoms; allergy involves multiple systems.
4 — B — sIgA neutralises toxins, viruses and bacteria and blocks adherence and antigen uptake.
5 — A — Altered flora lets *C. difficile* flourish.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam